

Module 2: Quantum Teleportation

Quantum Computing Workshop

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Quantum Teleportation: Overview

- Transfers quantum state from one qubit to another
- Requires:
 - 2 classical bits of communication
 - 1 pre-shared entangled qubit pair (Bell state)
- Does NOT transmit information faster than light
- First proposed by Bennett et al. (1993)
- Experimentally demonstrated in various quantum systems

Quantum Teleportation: Intuition

- Goal: Transfer unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ from Alice to Bob
- Cannot directly measure $|\psi\rangle$ (would destroy the state)
- Cannot copy $|\psi\rangle$ (no-cloning theorem)
- Solution: Use quantum entanglement as a resource
- The state is "teleported" without physical transfer of the qubit

No-Cloning Theorem

- Fundamental limitation in quantum mechanics
- It is impossible to create an identical copy of an arbitrary unknown quantum state
- For a hypothetical cloning operator U :

$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle \quad (1)$$

(2)

- Linearity of quantum mechanics prohibits this operation
- Classical information can be copied, quantum information cannot

Teleportation Protocol: Participants and Resources

- **Alice:** Has qubit in unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- **Bob:** Will receive the teleported state
- **Resources:**
 - Pre-shared Bell pair: $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
 - Classical communication channel (2 bits)
- **Initial system state:** $|\psi\rangle \otimes |\Phi^+\rangle$
- Alice holds two qubits (her original qubit and one from the Bell pair)
- Bob holds one qubit (the other half of the Bell pair)

Teleportation Protocol: Step 1

Alice applies a CNOT gate

- Control: Alice's state qubit
- Target: Alice's entangled qubit

$$\text{Initial: } |\psi\rangle_A \otimes \frac{1}{\sqrt{2}}(|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B) \quad (3)$$

$$= \frac{1}{\sqrt{2}}[\alpha |0\rangle_A (|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B) + \beta |1\rangle_A (|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B)] \quad (4)$$

$$\text{After CNOT: } \frac{1}{\sqrt{2}}[\alpha |0\rangle_A (|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B) \quad (5)$$

$$+ \beta |1\rangle_A (|1\rangle_A |0\rangle_B + |0\rangle_A |1\rangle_B)] \quad (6)$$

Teleportation Protocol: Step 2

Alice applies a Hadamard gate to her state qubit

Results in a superposition:

$$\text{After H: } \frac{1}{2}[\alpha(|0\rangle_A + |1\rangle_A)(|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B) + \quad (7)$$

$$\beta(|0\rangle_A - |1\rangle_A)(|1\rangle_A |0\rangle_B + |0\rangle_A |1\rangle_B)] \quad (8)$$

Key insight: The system is now in a superposition of 4 terms, each corresponding to a different measurement outcome and different state of Bob's qubit

Teleportation Protocol: Step 3

Alice measures both of her qubits

Four possible measurement outcomes:

- $|00\rangle$: Bob's qubit is in state $\alpha|0\rangle + \beta|1\rangle$
- $|01\rangle$: Bob's qubit is in state $\alpha|1\rangle + \beta|0\rangle$ (X correction needed)
- $|10\rangle$: Bob's qubit is in state $\alpha|0\rangle - \beta|1\rangle$ (Z correction needed)
- $|11\rangle$: Bob's qubit is in state $\alpha|1\rangle - \beta|0\rangle$ (XZ correction needed)

Alice sends her measurement result (2 classical bits) to Bob

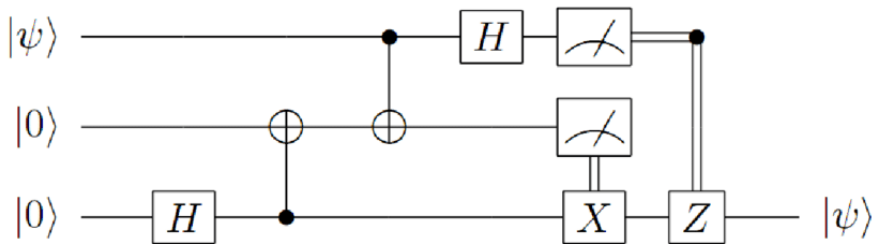
Teleportation Protocol: Step 4

Bob applies correction based on classical bits

Measurement	Correction Operation	Matrix
$ 00\rangle$	I (Identity)	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
$ 01\rangle$	X (Bit-flip)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
$ 10\rangle$	Z (Phase-flip)	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
$ 11\rangle$	ZX (Both)	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

After correction, Bob's qubit is guaranteed to be in the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

Quantum Teleportation Circuit



- Time flows from left to right
- Double lines from measurements indicate classical communication

Why "Teleportation"?

- The original qubit's state is destroyed in the process
- The state appears at Bob's location
- No physical particle travels from Alice to Bob
- The quantum information is transferred
- No-cloning theorem is preserved
- Requires classical communication (limited by speed of light)

Applications of Quantum Teleportation

- Foundational protocol in quantum information science
- Essential component for quantum networks
 - Enables secure transmission of quantum states
 - Addresses the challenge of transmitting qubits over long distances
- Basis for more complex quantum communication protocols

Key Takeaways

- Quantum teleportation transfers quantum states using entanglement
- Requires 2 classical bits + 1 entangled qubit pair
- Does NOT allow faster-than-light communication
- Process:
 - 1 Prepare entangled Bell pair
 - 2 Alice: CNOT + Hadamard + Measurement
 - 3 Classical communication of 2 bits
 - 4 Bob: Conditional correction (I, X, Z, or XZ)
- Demonstrates fundamental principles of quantum information

Next Steps

- Implementing quantum teleportation in Qiskit
- Introduction to quantum algorithms
- Deutsch's algorithm: the first quantum algorithm